Mental Workload EEG Database (STEW: Simultaneous Task EEG Workload Dataset)
About the Dataset

Original Database Link:
https://ieee-dataport.org/open-access/stew-simultaneous-task-eeg-workload-dataset

Please cite:

Wei Lun Lim, Olga Sourina, Lipo Wang, July 10, 2018, "STEW: Simultaneous Task EEG Workload Dataset", IEEE Dataport, doi: https://dx.doi.org/10.21227/44r8-ya50.

U. Singh and M. K. Ahirwal, "Mental Workload Classification for Multitasking Test using Electroencephalogram Signal," 2021 IEEE International Conference on Technology, Research, and Innovation for Betterment of Society (TRIBES), Raipur, India, 2021, pp. 1-6, doi: 10.1109/TRIBES52498.2021.9751676.

Utpal Singh and Mitul Kumar Ahirwal, Improved Classification of Mental Workload Using One Dimensional Convolutional Neural Network and SMOTE Technique, 7th International Conference on Graphics and Signal Processing at Fujisawa, Japan, June-2023. https://dl.acm.org/doi/10.1145/3606283.3606291

STEW: Simultaneous Task EEG Workload Data Set

This dataset consists of raw EEG data from 48 subjects who participated in a multitasking workload experiment utilizing the SIMKAP multitasking test. The subjects' brain activity at rest was also recorded before the test and is included as well. The Emotiv EPOC device, with sampling frequency of 128 Hz and 14 channels was used to obtain the data, with 2.5 minutes of EEG recording for each case. Subjects were also asked to rate their perceived mental workload after each stage on a rating scale of 1 to 9 and the ratings are provided in a separate file.

Uploaded Database on Keggle (in .mat format)

This database is of EEG (brain waves) signals, recorded when persons (subjects) were doing arithmetic calculations. This dataset aims to identify the levels of mental workload. Subjects also given a rating for mental workload felt by them (0 to 9 as difficulty level that indicates mental workload). So it is a supervised classification problem.

Experiment Design:

In the original database a total of 48 subjects are there. Note that ratings for subjects 5, 24 and 42 are unavailable. We do not include these subjects in study. Hence, only 45 subject's EEG data is available. For each subject, 14 channels recording are there (simply considered as time domain signals), for each signal 19200 values/ data points are there.

You may see it as a matrix of 14 x 19200 for each subject, treating it as one sample/ instant/ record of the database (hence, this database consists of a total 45 records).

dataset.mat = 14x19200x45
class_012.mat = 45x1
rating.mat=45x1
three_class_one_hot.mat=45x3

For classification we have to make two studies: one with 2 classes and another with 3 classes.

Three class classification: Class0: Normal Workload (4 5)
Class1: Moderate Workload (6 7)
Class2: High Workload (8 9)

Two class classification: Class0: Normal workload (rated as 4 5 6)
Class1: High workload (rated as 7 8 9)

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